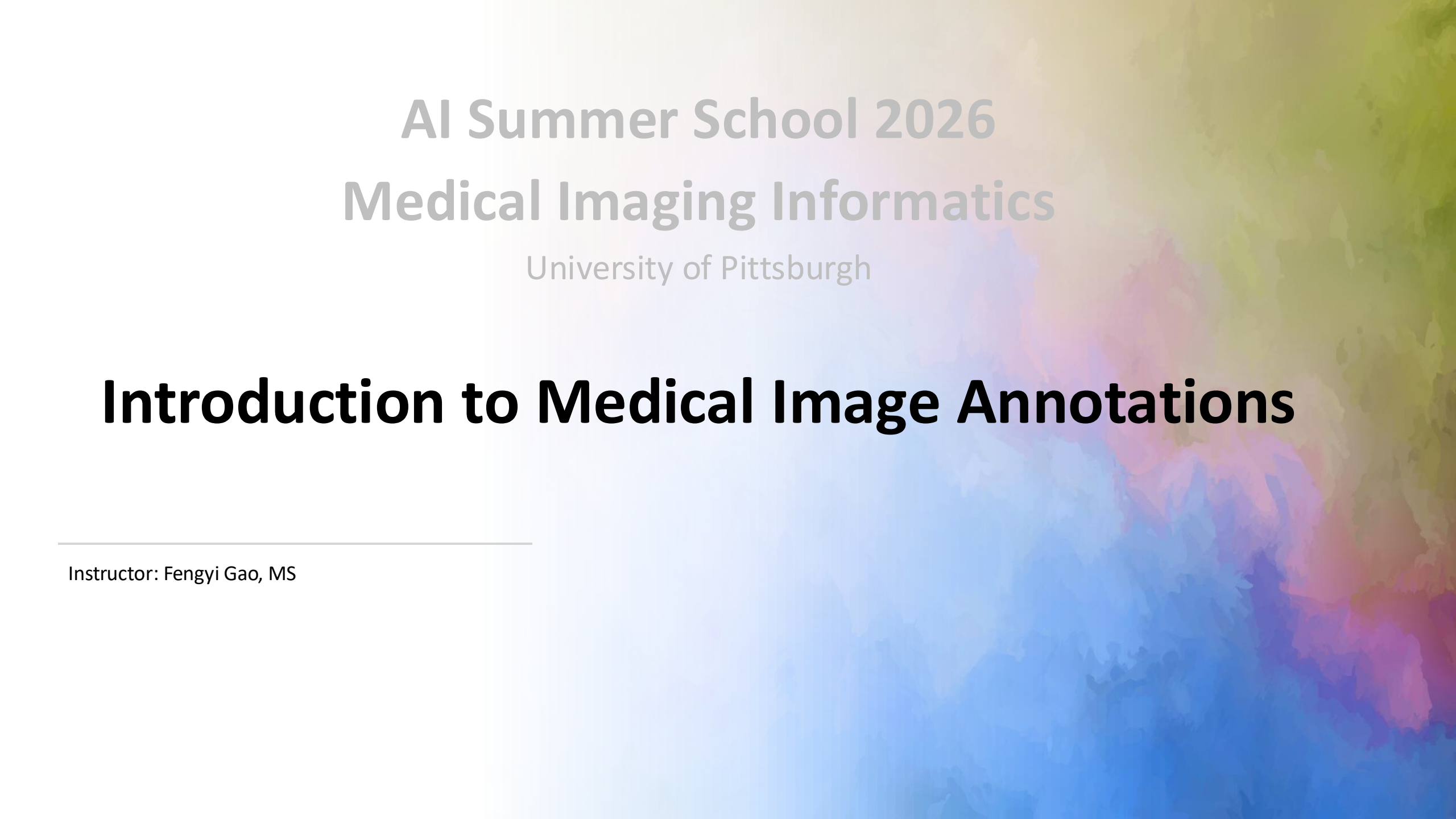




AI Summer School 2026

Medical Imaging Informatics

University of Pittsburgh

Introduction to Medical Image Annotations

Instructor: Fengyi Gao, MS

Learning Objectives

After completing this lecture, you should be able to:

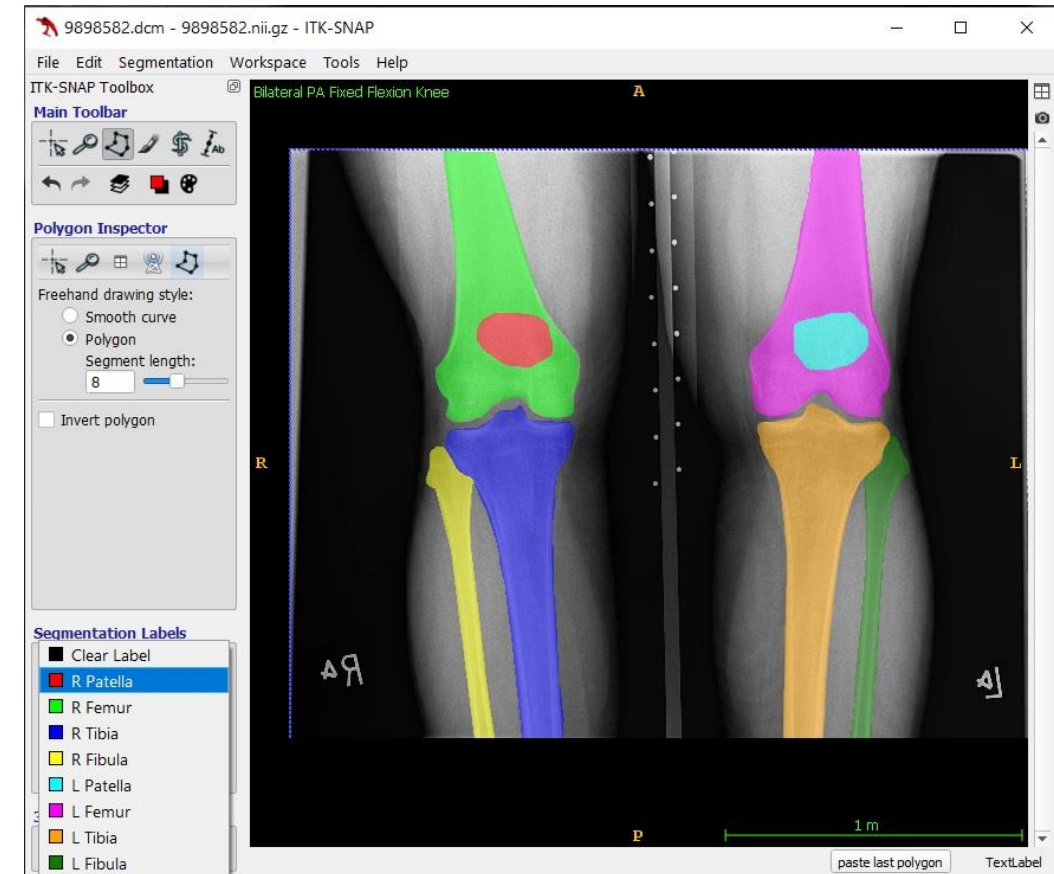
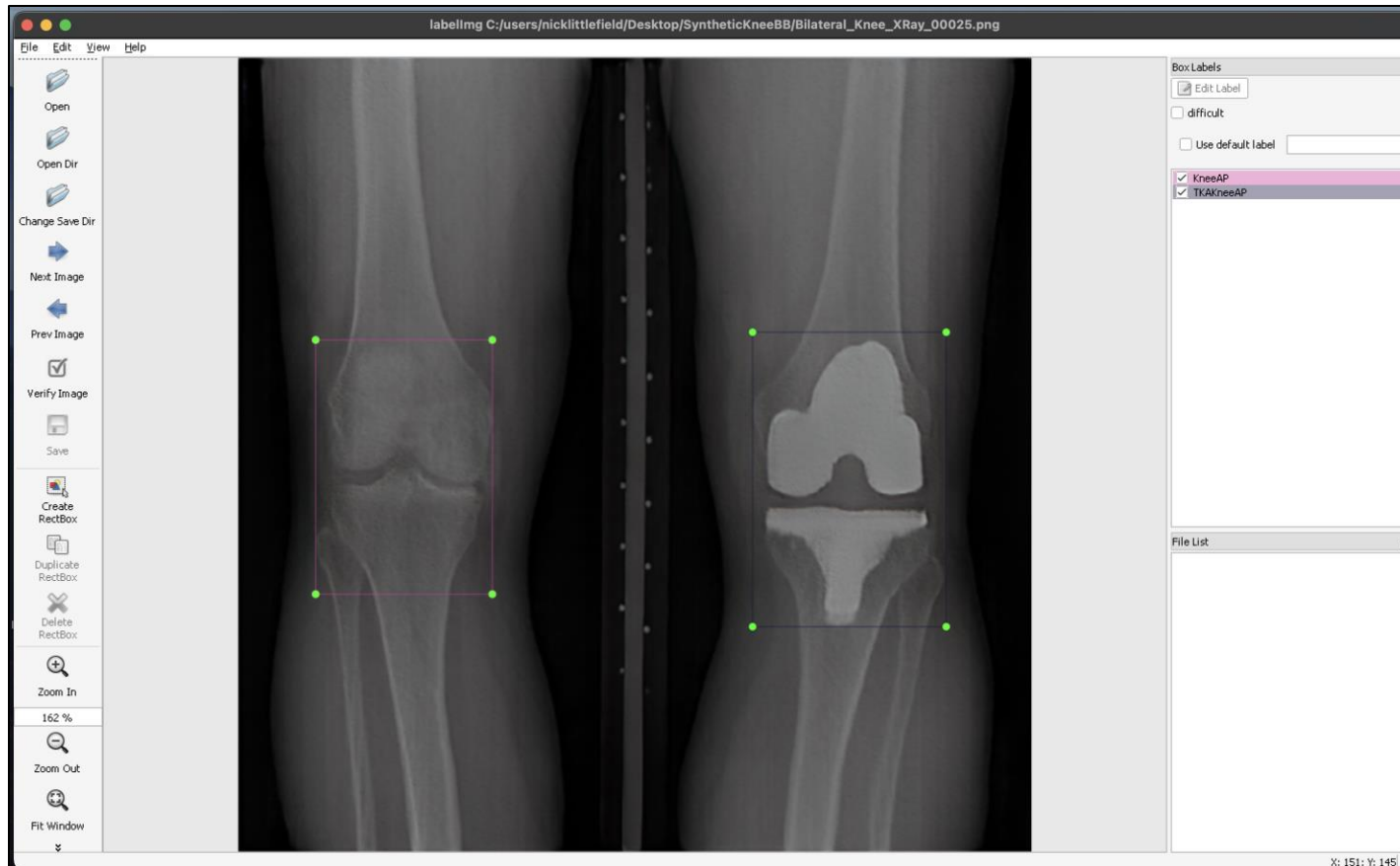
- Explain the purpose of medical image annotation
- Identify common annotation types in medical imaging
- Understand the typical annotation workflow
- Describe how annotation quality is assessed
- Recognize common annotation tools and challenges

Outline

- What is Medical Image Annotation?
- Types of Annotations
- Annotation Pipeline
- Annotation Quality Assessment
- Challenges
- Annotation Tools
- Common Errors
- Hands-on Practice

Medical Image Annotation: What and Why?

- **Medical image annotation** is the process of labeling medical images to identify structures, abnormalities, or regions of interest.



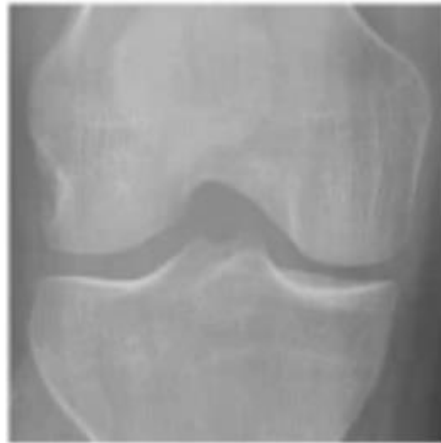
Types of Medical Image Annotation

- There are multiple types of medical image annotations:
 - Classification
 - Object Detection
 - Segmentation

Types of Medical Image Annotation: Classification

- Assigns a label to an image or region.
- Examples:
 - Benign vs. Malignant brain tumors
 - Lung cancer screening – benign, pre-cancerous, malignant,
 - Fracture types – simple, compound, ...

Image



Osteoarthritis (OA)

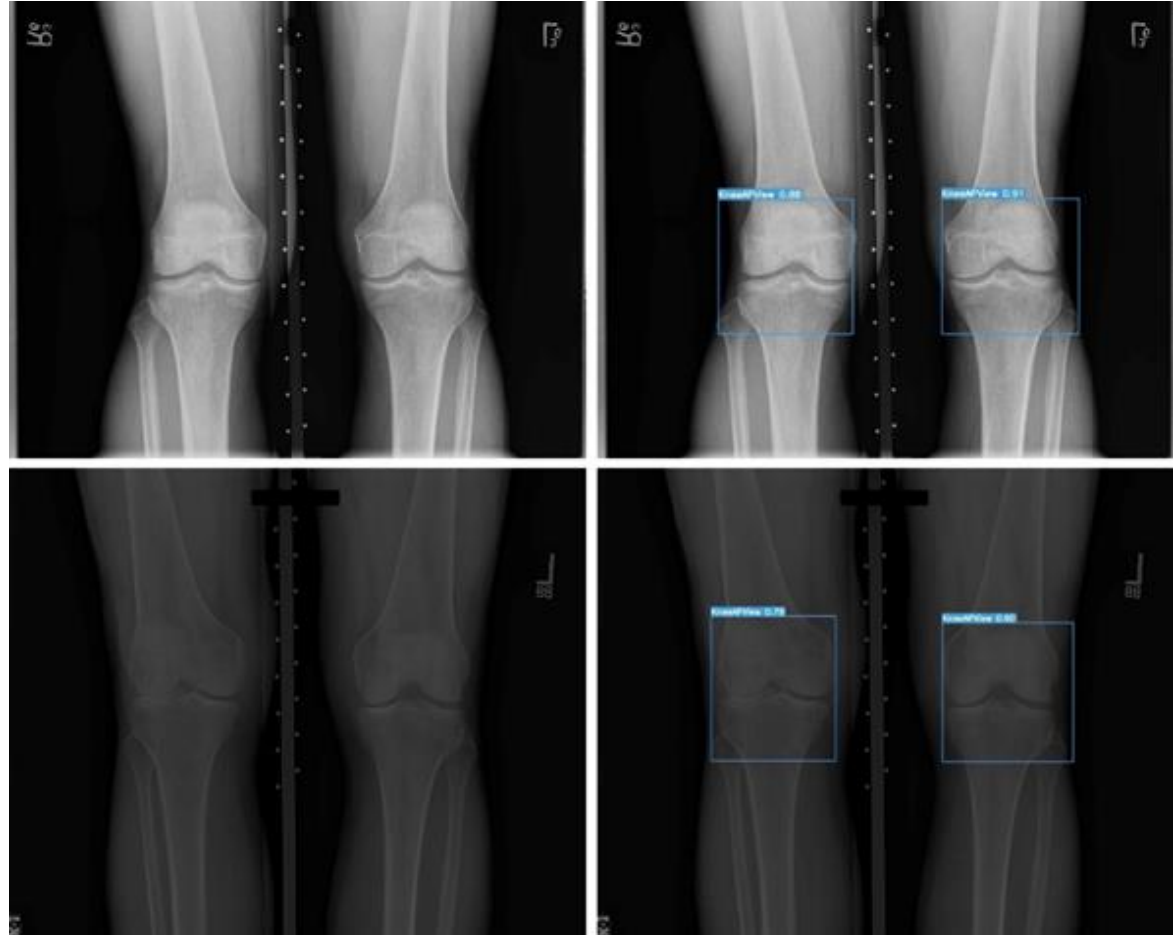


NON - Osteoarthritis (OA)

Label

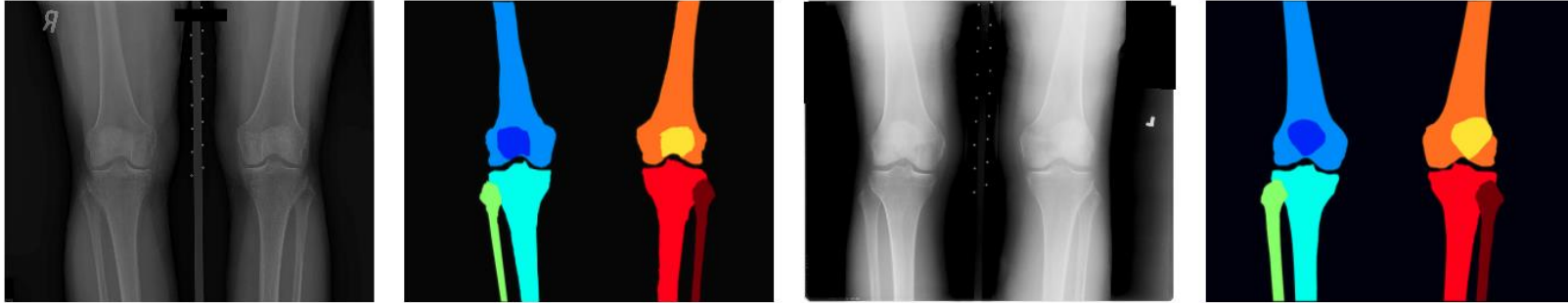
Types of Medical Image Annotation: Object Detection

- Identifies and localizes objects of interest using bounding boxes.
- Examples:
 - Knee joint area,
 - Brain tumors
 - Lung nodules
 - Cells in pathology images

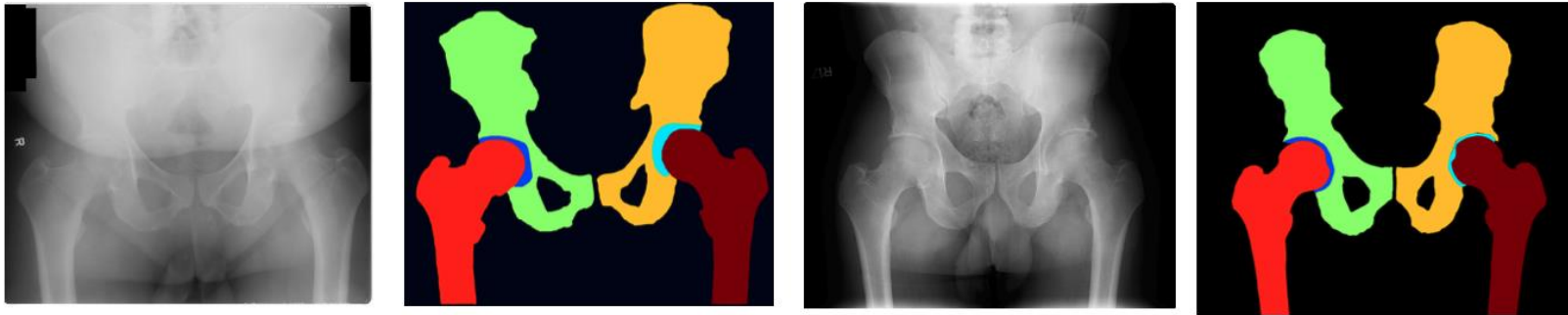


Types of Medical Image Annotation: Segmentation

- Outlines the exact boundaries of structures. (Organs, Tumors, Bones, Blood vessels)

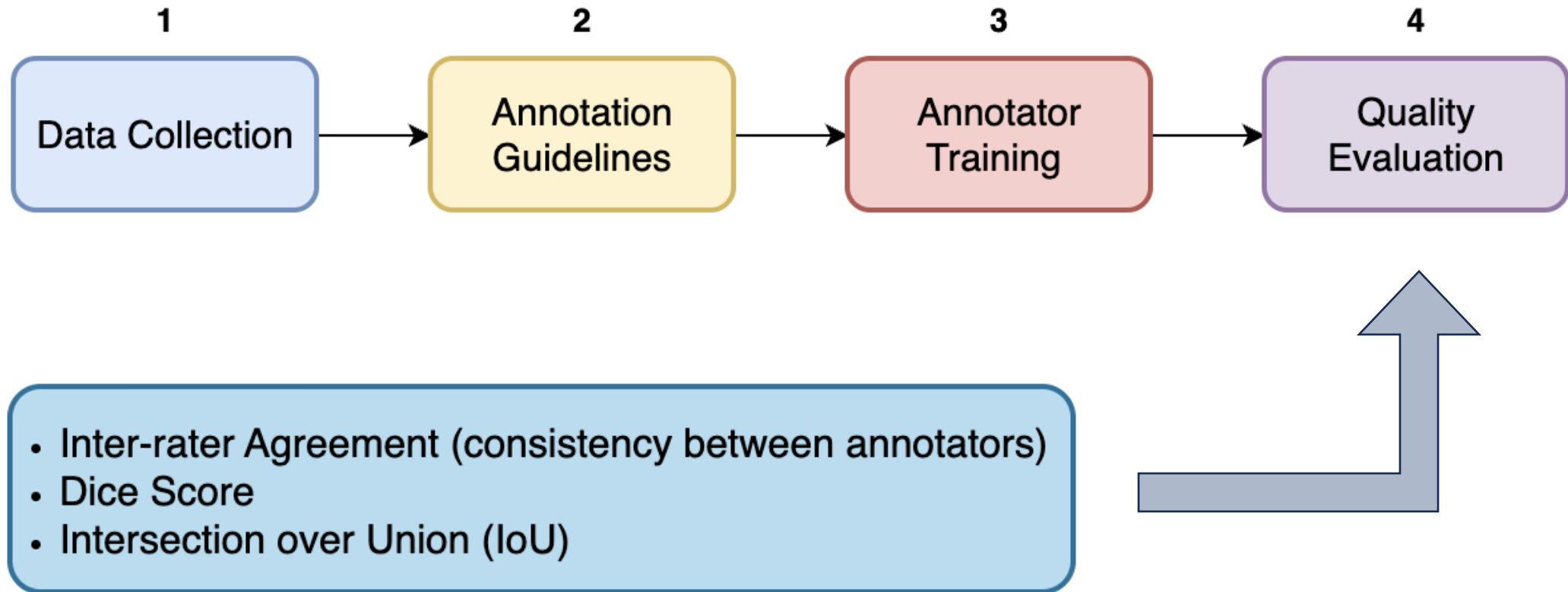


(a)



(b)

Typical Annotation Pipeline



Annotation Quality Assessment

- **Inter-rater Agreement**
 - Measures consistency between multiple annotators
 - Indicates how reliable the annotations are
 - Low agreement may suggest ambiguous cases or unclear guidelines

- **Annotation Quality Metrics**

- **Intersection over Union (IoU)**

- Measures overlap between two annotated regions
 - Range: 0 to 1
 - Higher IoU indicates greater agreement

- **Dice Score**

- Measures similarity between two annotated regions
 - Range: 0 to 1
 - 1 indicates perfect overlap

Diagram illustrating the Dice Score formula with labels:

- Overlapping region (points to $A \cap B$)
- Area (or pixels) in annotation A (points to $|A|$)
- Area (or pixels) in annotation B (points to $|B|$)
- Total area of Annotation A and Annotation B (points to $|A| + |B|$)

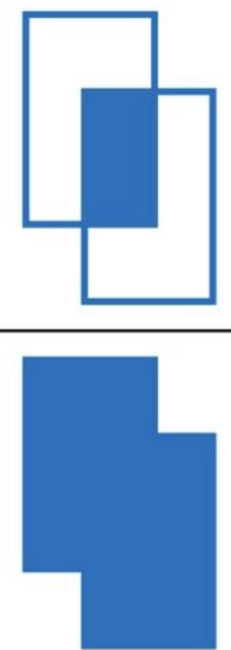
$$Dice = \frac{2|A \cap B|}{|A| + |B|}$$

Diagram illustrating the Intersection over Union (IoU) formula with labels:

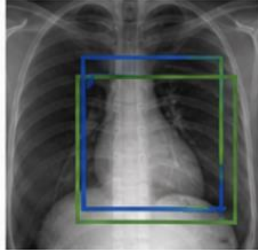
- Annotation A (points to the left rectangle)
- Annotation B (points to the right rectangle)
- Overlapping region (points to the intersection of the two rectangles)
- Union region (points to the combined area of both rectangles)

$$IoU = \frac{|A \cap B|}{|A \cup B|}$$

Score = $\frac{\text{Area of overlap}}{\text{Area of union}}$



How Agreement and Metrics Are Used Across Annotation Tasks

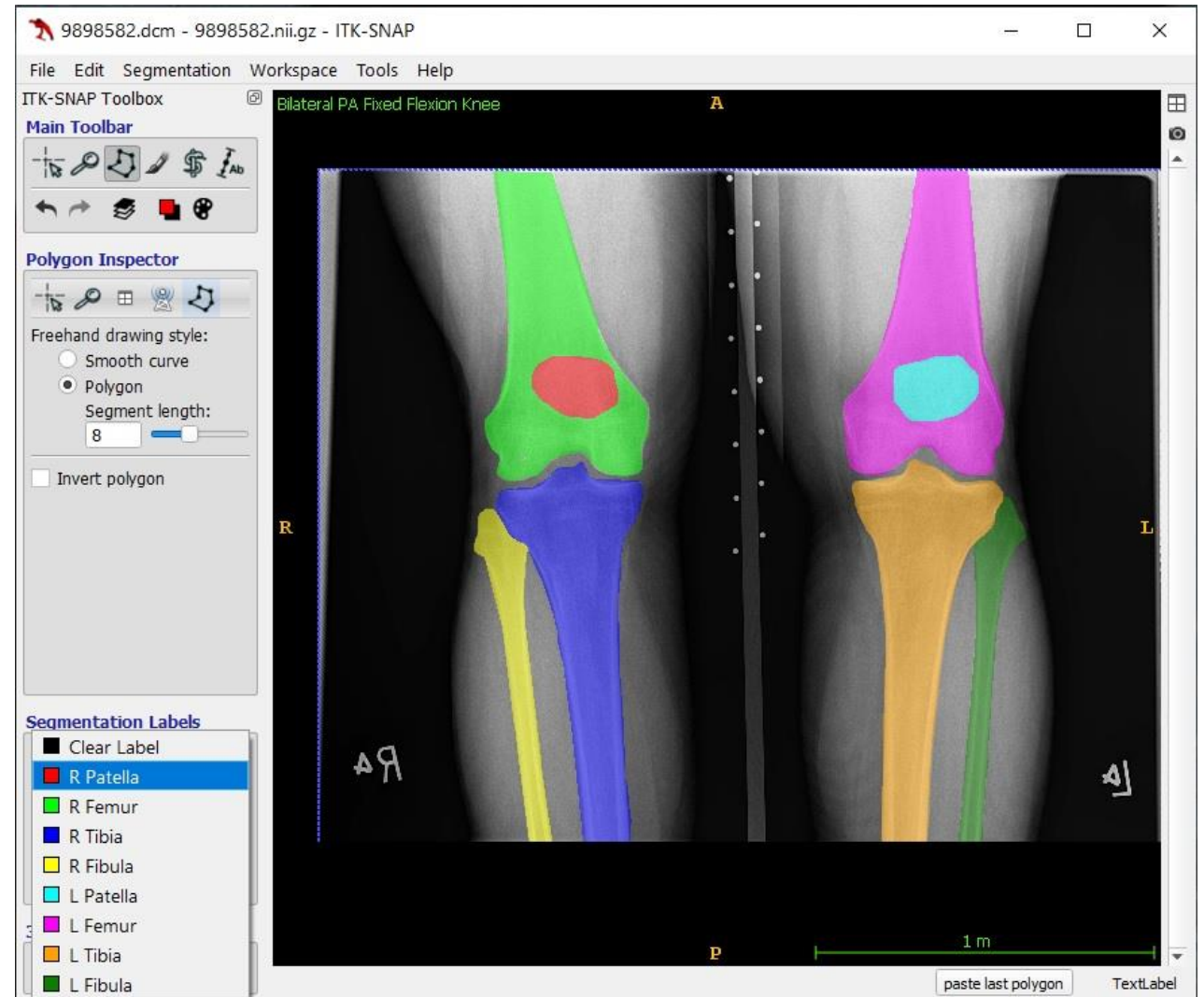
Task	Annotation Type	What Is Compared?	Example
Classification	Image-level label A single label is assigned to the entire image.	Label agreement Annotator A label vs Annotator B label	 Annotator A Annotator B Label: Pneumonia = Label: Pneumonia Agreement = Yes
Object Detection	Bounding Box Bounding boxes are drawn around objects of interest.	IoU on boxes Bounding Box A vs Bounding Box B	 Bounding Box A (Annotator A / Ground Truth) Bounding Box B (Annotator B / Model Prediction) Metric: IoU Higher IoU → better agreement
Segmentation	Pixel Mask Pixel-level masks are drawn to outline structures.	IoU / Dice on masks Segmentation Mask A vs Segmentation Mask B	Mask A (Annotator A / Ground Truth) Mask B (Annotator B / Model Prediction)  Metrics: • IoU • Dice Score Higher value → better agreement

Challenges with Medical Image Annotations

- Annotation accuracy
 - Annotations need to be precise and highly accurate to train AI models
- Annotation consistency
 - Different annotators can have varying interpretation of the same image
- Time and cost requirements
 - Detailed annotations take extensive amount of time
- Complexity of medical images
 - Images have different modalities, quality, and can contain artifacts

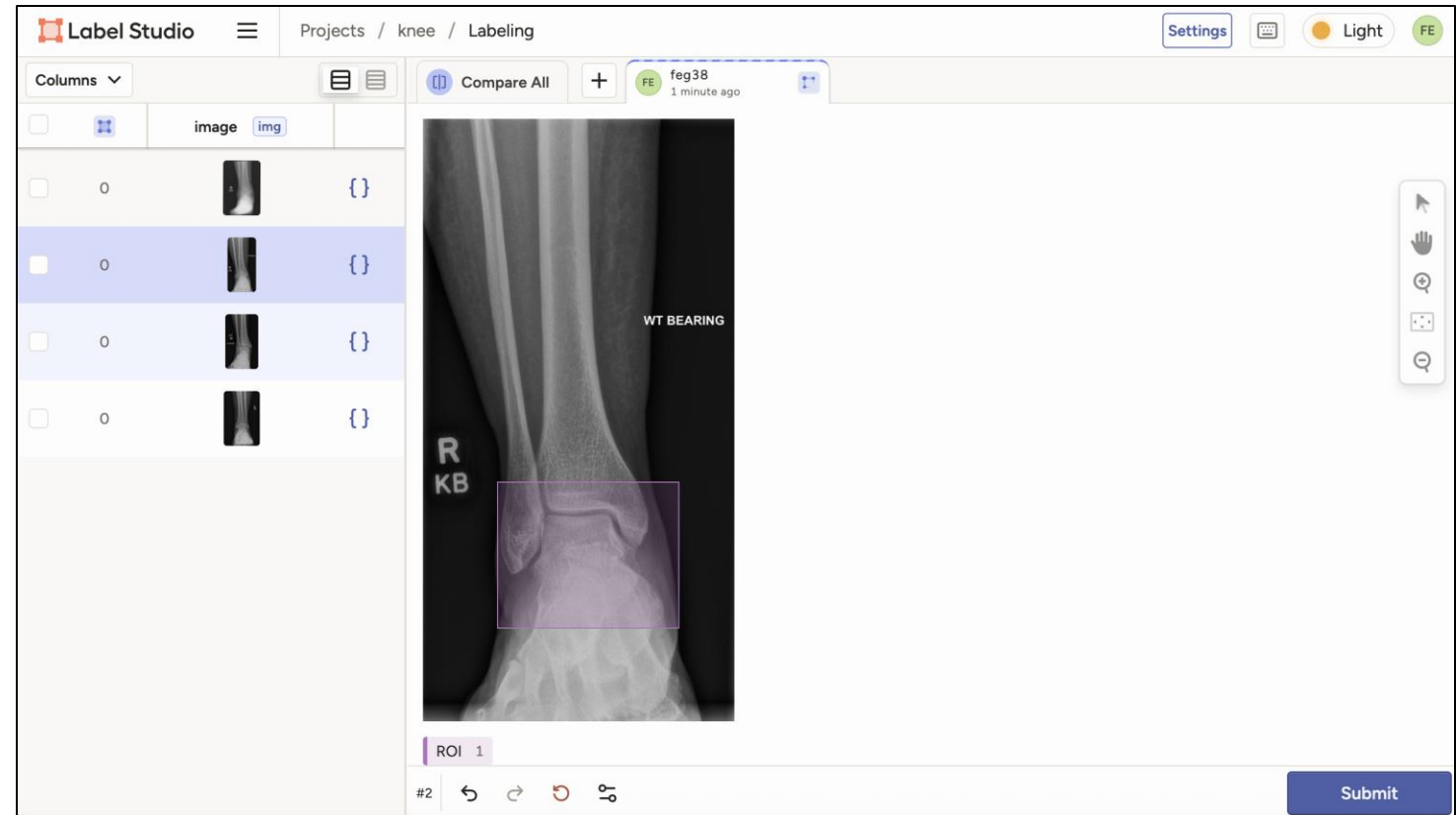
Tools for Object Detection Annotations: ITK-SNAP

- Open-source software for medical image segmentation
- Supports CT, MRI, and 3D medical images
- Allows manual and semi-automatic segmentation
- Commonly used in research and clinical studies



Tools for Object Detection Annotations: Label Studio

- Modern and user-friendly annotation platform for object detection
- Allows annotators to draw bounding boxes around objects of interest
- Supports image classification, object detection, and segmentation tasks
- Automatically stores object labels and bounding box coordinates
- Export annotations in common formats such as COCO, YOLO, and Pascal VOC



Common Mistakes in Image Annotation

- **Inaccurate Bounding Boxes**
 - Boxes are too loose or too tight
 - May reduce model accuracy
- **Inconsistent Labeling**
 - Different annotators use different labels. Example: "tumor" vs. "mass"
- **Missing Annotations**
 - Objects are not annotated
 - Lowers overall model performance.
- **Incorrect Class Labels**
 - Wrong category assigned. Example: benign labeled as malignant
- **Redundant Annotations**
 - Multiple boxes for the same object
 - May confuse the model
- **Image Artifacts**
 - Noise or imaging artifacts labeled as real structures
- **Non-Standard Formats**
 - Incorrect or incompatible annotation formats, such as YOLO, Pascal VOC, and COCO

Hand-On Practice: Manual Annotations in ITK-SNAP & Label Studio

- **ITK-SNAP** Link: <https://www.itksnap.org/pmwiki/pmwiki.php>
- **Label Studio** Link: <https://labelstud.io/>

Thank you!

Questions!



GitHub



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